FlagCX案例讲解

Case Study

时 间：2025.08.16

职 位：智源AI框架研究员

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

目录

FlagCX跨芯通信

测试机器

- 英伟达A800：10. 1. 15. 233

	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	NIC0	NIC1	NIC2	NIC3	NIC4	NIC5	NIC6	NIC7	NIC8	NIC9
GPU0	X	NV8	NV8	NV8	NV8	NV8	NV8	NV8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS
GPU1	NV8	X	NV8	NV8	NV8	NV8	NV8	NV8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS
GPU2	NV8	NV8	X	NV8	NV8	NV8	NV8	NV8	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS
GPU3	NV8	NV8	NV8	X	NV8	NV8	NV8	NV8	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS
GPU4	NV8	NV8	NV8	NV8	X	NV8	NV8	NV8	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB
GPU5	NV8	NV8	NV8	NV8	NV8	X	NV8	NV8	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB
GPU6	NV8	NV8	NV8	NV8	NV8	NV8	X	NV8	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS
GPU7	NV8	NV8	NV8	NV8	NV8	NV8	NV8	X	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS
NIC0	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS
NIC1	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS
NIC2	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS	SYS	SYS
NIC3	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS	SYS	SYS
NIC4	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS
NIC5	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS
NIC6	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	X	PXB	SYS	SYS
NIC7	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	PXB	X	SYS	SYS
NIC8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX
NIC9	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X

```
Legend:

  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  NV#  = Connection traversing a bonded set of # NVLinks

NIC Legend:

  NIC0: mlx5_0
  NIC1: mlx5_1
  NIC2: mlx5_101
  NIC3: mlx5_102
  NIC4: mlx5_103
  NIC5: mlx5_104
  NIC6: mlx5_105
  NIC7: mlx5_106
  NIC8: mlx5_107
  NIC9: mlx5_108
```

- 沐曦MC550：10. 1. 15. 67

```
mx-smi  version: 2.1.12

=================== MetaX System Management Interface Log ===================
Timestamp                                         : Fri Aug 15 17:48:26 2025

Attached GPUs                                     : 8
Device link type matrix
```

	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	Node Affinity	CPU Affinity
GPU0	X	MX	MX	MX	MX	MX	MX	MX	0	0-31
GPU1	MX	X	MX	MX	MX	MX	MX	MX	0	0-31
GPU2	MX	MX	X	MX	MX	MX	MX	MX	0	0-31
GPU3	MX	MX	MX	X	MX	MX	MX	MX	0	0-31
GPU4	MX	MX	MX	MX	X	MX	MX	MX	1	32-63
GPU5	MX	MX	MX	MX	MX	X	MX	MX	1	32-63
GPU6	MX	MX	MX	MX	MX	MX	X	MX	1	32-63
GPU7	MX	MX	MX	MX	MX	MX	MX	X	1	32-63

```
Legend:
  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  MX   = Connection traversing MetaXLink
  NA   = Connection type is unknown
```

Tests

Tests for FlagCX are maintained in `test/perf`.

```
cd test/perf
make [USE_NVIDIA/USE_ILUVATAR_COREX/USE_CAMBRICON/USE_METAX/USE_MUSA/USE_KUNLUNXIN/USE_DU/USE_
mpirun --allow-run-as-root -np 8 ./test_allreduce -b 128K -e 4G -f 2
```

Note that the default MPI install path is set to `/usr/local/mpi`, you may specify the MPI path with:

```
make MPI_HOME=<path to mpi install>
```

All tests support the same set of arguments:

- Sizes to scan
  - `-b <min size in bytes>` minimum size to start with. Default: 1M.
  - `-e <max size in bytes>` maximum size to end at. Default: 1G.
  - `-f <increment factor>` multiplication factor between sizes. Default: 2.
- Performance
  - `-w, <warmup iteration count>` number of warmup iterations (not timed). Default: 5.
  - `-n, <iteration count>` number of iterations. Default: 20.
- Utils
  - `-p, <0/1>` print buffer info. Default: 0.
  - `-h` print help message. Default: disabled.

FlagCX跨芯通信

测试机器

□ 英伟达A800：10. 1. 15. 233

	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	NIC0	NIC1	NIC2	NIC3	NIC4	NIC5	NIC6	NIC7	NIC8	NIC9
GPU0	X	NV8	NV8	NV8	NV8	NV8	NV8	NV8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS
GPU1	NV8	X	NV8	NV8	NV8	NV8	NV8	NV8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS
GPU2	NV8	NV8	X	NV8	NV8	NV8	NV8	NV8	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS
GPU3	NV8	NV8	NV8	X	NV8	NV8	NV8	NV8	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS
GPU4	NV8	NV8	NV8	NV8	X	NV8	NV8	NV8	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB
GPU5	NV8	NV8	NV8	NV8	NV8	X	NV8	NV8	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB
GPU6	NV8	NV8	NV8	NV8	NV8	NV8	X	NV8	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS
GPU7	NV8	NV8	NV8	NV8	NV8	NV8	NV8	X	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS
NIC0	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS
NIC1	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS
NIC2	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS	SYS	SYS
NIC3	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS	SYS	SYS
NIC4	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX	SYS	SYS	SYS	SYS
NIC5	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X	SYS	SYS	SYS	SYS
NIC6	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	X	PXB	SYS	SYS
NIC7	SYS	SYS	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	PXB	X	SYS	SYS
NIC8	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	X	PIX
NIC9	SYS	SYS	SYS	SYS	PXB	PXB	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	SYS	PIX	X

```
Legend:

  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  NV#  = Connection traversing a bonded set of # NVLinks

NIC Legend:

  NIC0: mlx5_0
  NIC1: mlx5_1
  NIC2: mlx5_101
  NIC3: mlx5_102
  NIC4: mlx5_103
  NIC5: mlx5_104
  NIC6: mlx5_105
  NIC7: mlx5_106
  NIC8: mlx5_107
  NIC9: mlx5_108
```

□ 沐曦MC550：10. 1. 15. 67

```
mx-smi  version: 2.1.12

=================== MetaX System Management Interface Log ===================
Timestamp                                      : Fri Aug 15 17:48:26 2025

Attached GPUs                                  : 8
Device link type matrix
```

	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	Node Affinity	CPU Affinity
GPU0	X	MX	MX	MX	MX	MX	MX	MX	0	0-31
GPU1	MX	X	MX	MX	MX	MX	MX	MX	0	0-31
GPU2	MX	MX	X	MX	MX	MX	MX	MX	0	0-31
GPU3	MX	MX	MX	X	MX	MX	MX	MX	0	0-31
GPU4	MX	MX	MX	MX	X	MX	MX	MX	1	32-63
GPU5	MX	MX	MX	MX	MX	X	MX	MX	1	32-63
GPU6	MX	MX	MX	MX	MX	MX	X	MX	1	32-63
GPU7	MX	MX	MX	MX	MX	MX	MX	X	1	32-63

```
Legend:
  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  MX   = Connection traversing MetaXLink
  NA   = Connection type is unknown
```

□ A800

```
docker start -ia flagcx-changtao
cd /root/FlagCX/
MPI_HOME=/workspace/mpich-4.2.3/build
CCL_HOME=/workspace/nccl/build/ USE_NVIDIA=1 make -j32


cd test/perf
MPI_HOME=/workspace/mpich-4.2.3/build
CCL_HOME=/workspace/nccl/build/ USE_NVIDIA=1 make -j32
```

□ MC550

```
docker start -ia flagcx-changtao
cd /root/FlagCX/
make USE_METAX=1 MPI_HOME=/workspace/mpich-
4.2.3/build/ -j32

cd test/perf
make USE_METAX=1 MPI_HOME=/workspace/mpich-
4.2.3/build/ -j32
```

FlagCX跨芯通信

测试命令

A800

cd /root/FlagCX/test/perf

bash run_check.sh [setup.sh] [op_type]

测试一：常规测试

```
# Loop through each op type and run the test
for op in "${op_types[@]}"; do
  echo "[INFO] Launching test for op_type: $op"
  /root/mpich/build/bin/mpirun -np 16 -hosts 10.1.15.233:8,10.1.15.67:8 \
    -genv PATH=/root/mpich/build/bin:/opt/maca/bin:/usr/local/corex/bin \
    -genv LD_LIBRARY_PATH=/root/mpich/build/lib:/opt/maca/lib:/usr/local/corex/lib \
    -genv FLAGCX_ENABLE_TOPO_DETECT=TRUE \
    -genv FLAGCX_DEBUG=INFO \
    -genv FLAGCX_DEBUG_SUBSYS=INIT \
    /bin/bash /root/FlagCX/test/perf/${setup} "$op"
  echo "[INFO] Completed test for $op"
  echo "----------------------------------------"
done
```

测试二：流水优化测试

```
# Loop through each op type and run the test
for op in "${op_types[@]}"; do
  echo "[INFO] Launching pipeline test for op_type: $op"
  /root/mpich/build/bin/mpirun -np 16 -hosts 10.1.15.233:8,10.1.15.67:8 \
    -genv PATH=/root/mpich/build/bin:/opt/maca/bin:/usr/local/corex/bin \
    -genv LD_LIBRARY_PATH=/root/mpich/build/lib:/opt/maca/lib:/usr/local/corex/lib \
    -genv FLAGCX_ENABLE_TOPO_DETECT=TRUE \
    -genv FLAGCX_DEBUG=INFO \
    -genv FLAGCX_DEBUG_SUBSYS=INIT \
    -genv FLAGCX_C2C_ALGO=RING_PIPELINED \
    /bin/bash /root/FlagCX/test/perf/${setup} "$op"
  echo "[INFO] Completed pipeline test for $op"
  echo "----------------------------------------"
done
```

setup.sh

- setup1，8nic-8nic
- setup2，2nic-4nic
- setup3，2nic-3nic
- setup4，1nic-1nic

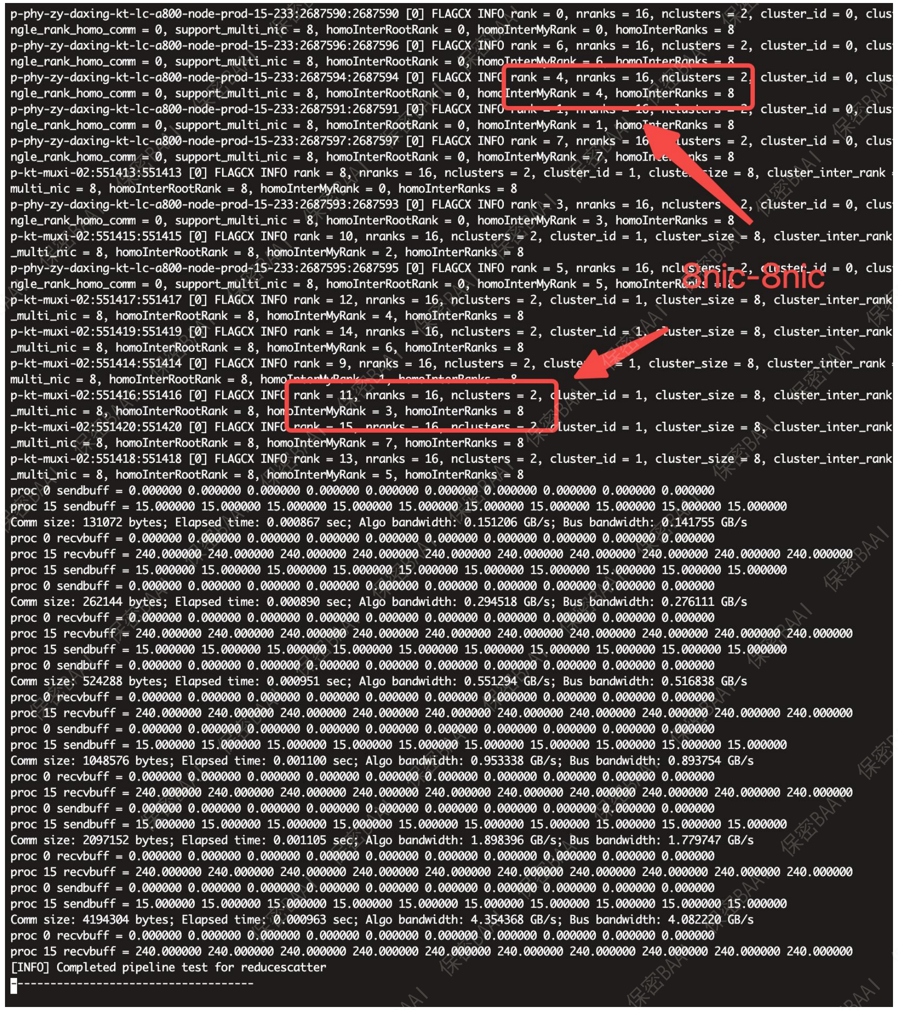
op_type

- sendrecv，alltoall，alltoallv
- broadcast，scatter，gather，allgather
- reduce，allreduce，，reducescatter

FlagCX跨芯通信

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

测试结果




```
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687590:2687590 [0] FLAGCX INFO rank = 0, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 0, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687596:2687596 [0] FLAGCX INFO rank = 6, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 6, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687594:2687594 [0] FLAGCX INFO rank = 4, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 4, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687591:2687591 [0] FLAGCX INFO [illegible] cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 1, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687597:2687597 [0] FLAGCX INFO rank = 7, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 7, homoInterRanks = 8
p-kt-muxi-02:551413:551413 [0] FLAGCX INFO rank = 8, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 0, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687593:2687593 [0] FLAGCX INFO rank = 3, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 3, homoInterRanks = 8
p-kt-muxi-02:551415:551415 [0] FLAGCX INFO rank = 10, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 2, homoInterRanks = 8
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2687595:2687595 [0] FLAGCX INFO rank = 5, nranks = 16, nclusters = 2, cluster_id = 0, clus
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 0, homoInterMyRank = 5, homoInterRanks = 8
p-kt-muxi-02:551417:551417 [0] FLAGCX INFO rank = 12, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 4, homoInterRanks = 8
p-kt-muxi-02:551419:551419 [0] FLAGCX INFO rank = 14, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 6, homoInterRanks = 8
p-kt-muxi-02:551414:551414 [0] FLAGCX INFO rank = 9, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
multi_nic = 8, homoInterRootRank = 8, [illegible]
p-kt-muxi-02:551416:551416 [0] FLAGCX INFO rank = 11, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 3, homoInterRanks = 8
p-kt-muxi-02:551420:551420 [0] FLAGCX INFO [illegible] cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 7, homoInterRanks = 8
p-kt-muxi-02:551418:551418 [0] FLAGCX INFO rank = 13, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 5, homoInterRanks = 8
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 131072 bytes; Elapsed time: 0.000867 sec; Algo bandwidth: 0.151206 GB/s; Bus bandwidth: 0.141755 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 262144 bytes; Elapsed time: 0.000890 sec; Algo bandwidth: 0.294518 GB/s; Bus bandwidth: 0.276111 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 524288 bytes; Elapsed time: 0.000951 sec; Algo bandwidth: 0.551294 GB/s; Bus bandwidth: 0.516838 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 1048576 bytes; Elapsed time: 0.001100 sec; Algo bandwidth: 0.953338 GB/s; Bus bandwidth: 0.893754 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 2097152 bytes; Elapsed time: 0.001105 sec; Algo bandwidth: 1.898396 GB/s; Bus bandwidth: 1.779747 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 4194304 bytes; Elapsed time: 0.000963 sec; Algo bandwidth: 4.354368 GB/s; Bus bandwidth: 4.082220 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
[INFO] Completed pipeline test for reducescatter
----------------------------------
```

A800 setup1.sh

代码块

```bash
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102,mlx5_103,mlx5_104,mlx5_105,mlx5_106,mlx5_107,mlx5_108

echo "[INFO] Running: test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1
```

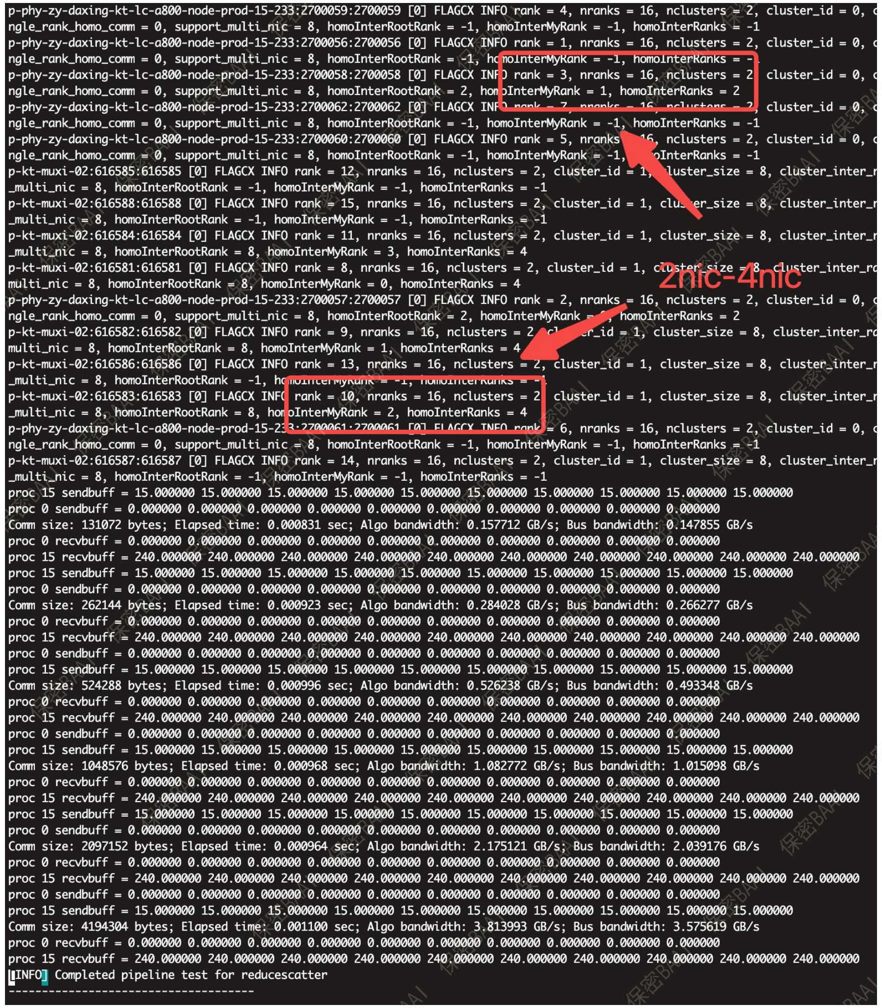
MC550 setup1.sh

代码块

```bash
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[INFO] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102,mlx5_103,mlx5_104,mlx5_105,mlx5_106,mlx5_107,mlx5_108

echo "[INFO] Running: test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1
```

FlagCX跨芯通信

测试结果



```
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700059:2700059 [0] FLAGCX INFO rank = 4, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700056:2700056 [0] FLAGCX INFO rank = 1, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700058:2700058 [0] FLAGCX INFO rank = 3, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 2, homoInterMyRank = 1, homoInterRanks = 2
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700062:2700062 [0] FLAGCX INFO rank = 7, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700060:2700060 [0] FLAGCX INFO rank = 5, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:616585:616585 [0] FLAGCX INFO rank = 12, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:616588:616588 [0] FLAGCX INFO rank = 15, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:616584:616584 [0] FLAGCX INFO rank = 11, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 3, homoInterRanks = 4
p-kt-muxi-02:616581:616581 [0] FLAGCX INFO rank = 8, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_ra
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 0, homoInterRanks = 4
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700057:2700057 [0] FLAGCX INFO rank = 2, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 2, homoInterMyRank = [illegible] homoInterRanks = 2
p-kt-muxi-02:616582:616582 [0] FLAGCX INFO rank = 9, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_ra
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 1, homoInterRanks = 4
p-kt-muxi-02:616586:616586 [0] FLAGCX INFO rank = 13, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:616583:616583 [0] FLAGCX INFO rank = 10, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 2, homoInterRanks = 4
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2700061:2700061 [0] FLAGCX INFO rank = 6, nranks = 16, nclusters = 2, cluster_id = 0, c
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:616587:616587 [0] FLAGCX INFO rank = 14, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_r
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 131072 bytes; Elapsed time: 0.000831 sec; Algo bandwidth: 0.157712 GB/s; Bus bandwidth: 0.147855 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 262144 bytes; Elapsed time: 0.000923 sec; Algo bandwidth: 0.284028 GB/s; Bus bandwidth: 0.266277 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 524288 bytes; Elapsed time: 0.000996 sec; Algo bandwidth: 0.526238 GB/s; Bus bandwidth: 0.493348 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 1048576 bytes; Elapsed time: 0.000968 sec; Algo bandwidth: 1.082772 GB/s; Bus bandwidth: 1.015098 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 2097152 bytes; Elapsed time: 0.000964 sec; Algo bandwidth: 2.175121 GB/s; Bus bandwidth: 2.039176 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 4194304 bytes; Elapsed time: 0.001100 sec; Algo bandwidth: 3.813993 GB/s; Bus bandwidth: 3.575619 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000 240.000000
[INFO] Completed pipeline test for reducescatter
------------------------------------
```


A800 setup2.sh

代码块

```
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102

echo "[INFO] Running: test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1
```

MC550 setup2.sh

代码块

```
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102,mlx5_103,mlx5_104

echo "[INFO] Running: test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1
```

FlagCX跨芯通信

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

测试结果

```
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809858:2809858 [0] FLAGCX INFO rank = 0, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809859:2809859 [0] FLAGCX INFO rank = 1, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809861:2809861 [0] FLAGCX INFO rank = 3, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 2, homoInterMyRank = 1, homoInterRanks = 2
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809863:2809863 [0] FLAGCX INFO rank = 5, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809864:2809864 [0] FLAGCX INFO rank = 6, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809865:2809865 [0] FLAGCX INFO rank = 7, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801455:801455 [0] FLAGCX INFO rank = 8, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 0, homoInterRanks = 3
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2809862:2809862 [0] FLAGCX INFO rank = 4, nranks = 16, nclusters = 2, cluster_i
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801456:801456 [0] FLAGCX INFO rank = 9, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 1, homoInterRanks = 3
p-kt-muxi-02:801457:801457 [0] FLAGCX INFO rank = 10, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 2, homoInterRanks = 3
p-kt-muxi-02:801458:801458 [0] FLAGCX INFO rank = 11, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801461:801461 [0] FLAGCX INFO rank = 14, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801460:801460 [0] FLAGCX INFO rank = 13, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801462:801462 [0] FLAGCX INFO rank = 15, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:801459:801459 [0] FLAGCX INFO rank = 12, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 131072 bytes; Elapsed time: 0.001443 sec; Algo bandwidth: 0.090802 GB/s; Bus bandwidth: 0.085127 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 262144 bytes; Elapsed time: 0.001274 sec; Algo bandwidth: 0.205730 GB/s; Bus bandwidth: 0.192872 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
Comm size: 524288 bytes; Elapsed time: 0.001323 sec; Algo bandwidth: 0.396185 GB/s; Bus bandwidth: 0.371424 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 1048576 bytes; Elapsed time: 0.001389 sec; Algo bandwidth: 0.754900 GB/s; Bus bandwidth: 0.707719 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 2097152 bytes; Elapsed time: 0.001655 sec; Algo bandwidth: 1.267499 GB/s; Bus bandwidth: 1.188280 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
Comm size: 4194304 bytes; Elapsed time: 0.001814 sec; Algo bandwidth: 2.312293 GB/s; Bus bandwidth: 2.167775 GB/s
proc 0 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 recvbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 0 sendbuff = 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
proc 15 sendbuff = 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000 15.000000
```

2nic-3nic

A800 setup3.sh

代码块

```
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102

echo "[INFO] Running: test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1
```

MC550 setup3.sh

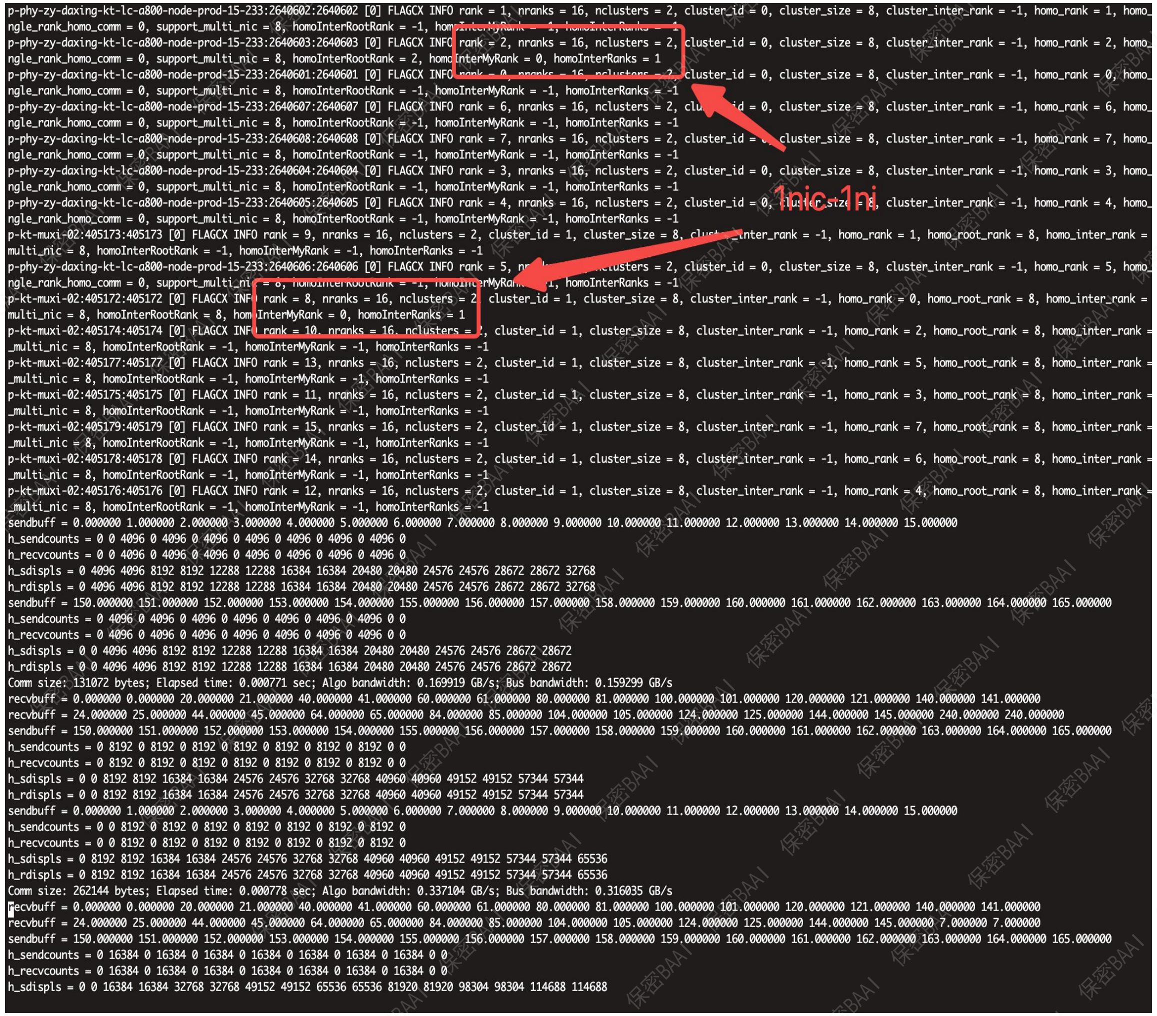
代码块

```
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101,mlx5_102,mlx5_103

echo "[INFO] Running: test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4G -f 2 -r 0 -w 5 -n 100 -p 1
```

FlagCX跨芯通信

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

测试结果




```
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640602:2640602 [0] FLAGCX INFO rank = 1, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 1, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640603:2640603 [0] FLAGCX INFO rank = 2, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 2, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = 2, homoInterMyRank = 0, homoInterRanks = 1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640601:2640601 [0] FLAGCX INFO rank = 0, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 0, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640607:2640607 [0] FLAGCX INFO rank = 6, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 6, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640608:2640608 [0] FLAGCX INFO rank = 7, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 7, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640604:2640604 [0] FLAGCX INFO rank = 3, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 3, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640605:2640605 [0] FLAGCX INFO rank = 4, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 4, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405173:405173 [0] FLAGCX INFO rank = 9, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 1, homo_root_rank = 8, homo_inter_rank =
multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-phy-zy-daxing-kt-lc-a800-node-prod-15-233:2640606:2640606 [0] FLAGCX INFO rank = 5, nranks = 16, nclusters = 2, cluster_id = 0, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 5, homo_
ngle_rank_homo_comm = 0, support_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405172:405172 [0] FLAGCX INFO rank = 8, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 0, homo_root_rank = 8, homo_inter_rank =
multi_nic = 8, homoInterRootRank = 8, homoInterMyRank = 0, homoInterRanks = 1
p-kt-muxi-02:405174:405174 [0] FLAGCX INFO rank = 10, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 2, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405177:405177 [0] FLAGCX INFO rank = 13, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 5, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405175:405175 [0] FLAGCX INFO rank = 11, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 3, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405179:405179 [0] FLAGCX INFO rank = 15, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 7, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405178:405178 [0] FLAGCX INFO rank = 14, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 6, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
p-kt-muxi-02:405176:405176 [0] FLAGCX INFO rank = 12, nranks = 16, nclusters = 2, cluster_id = 1, cluster_size = 8, cluster_inter_rank = -1, homo_rank = 4, homo_root_rank = 8, homo_inter_rank =
_multi_nic = 8, homoInterRootRank = -1, homoInterMyRank = -1, homoInterRanks = -1
sendbuff = 0.000000 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000 10.000000 11.000000 12.000000 13.000000 14.000000 15.000000
h_sendcounts = 0 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0
h_recvcounts = 0 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0
h_sdispls = 0 4096 4096 8192 8192 12288 12288 16384 16384 20480 20480 24576 24576 28672 28672 32768
h_rdispls = 0 4096 4096 8192 8192 12288 12288 16384 16384 20480 20480 24576 24576 28672 28672 32768
sendbuff = 150.000000 151.000000 152.000000 153.000000 154.000000 155.000000 156.000000 157.000000 158.000000 159.000000 160.000000 161.000000 162.000000 163.000000 164.000000 165.000000
h_sendcounts = 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 0
h_recvcounts = 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 4096 0 0
h_sdispls = 0 0 4096 4096 8192 8192 12288 12288 16384 16384 20480 20480 24576 24576 28672 28672
h_rdispls = 0 0 4096 4096 8192 8192 12288 12288 16384 16384 20480 20480 24576 24576 28672 28672
Comm size: 131072 bytes; Elapsed time: 0.000771 sec; Algo bandwidth: 0.169919 GB/s; Bus bandwidth: 0.159299 GB/s
recvbuff = 0.000000 0.000000 20.000000 21.000000 40.000000 41.000000 60.000000 61.000000 80.000000 81.000000 100.000000 101.000000 120.000000 121.000000 140.000000 141.000000
recvbuff = 24.000000 25.000000 44.000000 45.000000 64.000000 65.000000 84.000000 85.000000 104.000000 105.000000 124.000000 125.000000 144.000000 145.000000 240.000000 240.000000
sendbuff = 150.000000 151.000000 152.000000 153.000000 154.000000 155.000000 156.000000 157.000000 158.000000 159.000000 160.000000 161.000000 162.000000 163.000000 164.000000 165.000000
h_sendcounts = 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 0
h_recvcounts = 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 0
h_sdispls = 0 0 8192 8192 16384 16384 24576 24576 32768 32768 40960 40960 49152 49152 57344 57344
h_rdispls = 0 0 8192 8192 16384 16384 24576 24576 32768 32768 40960 40960 49152 49152 57344 57344
sendbuff = 0.000000 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000 10.000000 11.000000 12.000000 13.000000 14.000000 15.000000
h_sendcounts = 0 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0
h_recvcounts = 0 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0 8192 0
h_sdispls = 0 8192 8192 16384 16384 24576 24576 32768 32768 40960 40960 49152 49152 57344 57344 65536
h_rdispls = 0 8192 8192 16384 16384 24576 24576 32768 32768 40960 40960 49152 49152 57344 57344 65536
Comm size: 262144 bytes; Elapsed time: 0.000778 sec; Algo bandwidth: 0.337104 GB/s; Bus bandwidth: 0.316035 GB/s
recvbuff = 0.000000 0.000000 20.000000 21.000000 40.000000 41.000000 60.000000 61.000000 80.000000 81.000000 100.000000 101.000000 120.000000 121.000000 140.000000 141.000000
recvbuff = 24.000000 25.000000 44.000000 45.000000 64.000000 65.000000 84.000000 85.000000 104.000000 105.000000 124.000000 125.000000 144.000000 145.000000 7.000000 7.000000
sendbuff = 150.000000 151.000000 152.000000 153.000000 154.000000 155.000000 156.000000 157.000000 158.000000 159.000000 160.000000 161.000000 162.000000 163.000000 164.000000 165.000000
h_sendcounts = 0 16384 0 16384 0 16384 0 16384 0 16384 0 16384 0 16384 0 0
h_recvcounts = 0 16384 0 16384 0 16384 0 16384 0 16384 0 16384 0 16384 0 0
h_sdispls = 0 0 16384 16384 32768 32768 49152 49152 65536 65536 81920 81920 98304 98304 114688 114688
```

A800 setup4.sh

代码块

```bash
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101

echo "[INFO] Running: test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1
```

MC550 setup4.sh

代码块

```bash
#!/bin/bash

op_type="$1"

if [ -z "$op_type" ]; then
  echo "[ERROR] No op_type specified."
  echo "Usage: $0 [op_type]"
  exit 1
fi

export FLAGCX_IB_HCA=mlx5_101

echo "[INFO] Running: test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1"
exec /root/FlagCX/test/perf/test_${op_type} -b 128K -e 4M -f 2 -r 0 -w 5 -n 100 -p 1
```

FlagCX跨芯通信

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

测试结果

通信操作	多网卡						单网卡	
	8-8（setup1）		2-4（setup2）		2-3（setup3）		1-1（setup4）	
	w/ pipeline	w/o pipeline	w/ pipeline	w/o pipeline	w/ pipeline	w/o pipeline	w/ pipeline	w/o pipeline
sendrecv	-	✅	-	✅	-	✅	-	✅
broadcast	-	✅	-	✅	-	✅	-	✅
gather	-	✅	-	✅	-	✅	-	✅
scatter	-	✅	-	✅	-	✅	-	✅
reduce	-	✅	-	✅	-	✅	-	✅
allreduce	✅	✅	✅	✅	✅	✅	✅	✅
allgather	✅	✅	✅	✅	✅	✅	✅	✅
reducescatter	✅	✅	✅	✅	✅	✅	✅	✅
alltoall	-	✅	-	✅	-	✅	-	✅
alltoallv	-	✅	-	✅	-	✅	-	✅

FlagScale+FlagCX异构PD分离

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

FlagScale代

https://github.com/FlagOpen/FlagScale/blob/main/flagscale/backends/vllm/vllm/distributed/kv_transfer/kv_pipe/flagcx_p2p_nccl_pipe.py

FlagScale / flagscale / backends / vllm / vllm / distributed / kv_transfer / kv_pipe / **flagcx_p2p_nccl_pipe.py**

Code | Blame | 471 lines (414 loc) · 19.6 KB

```python

class P2pNcclPipe:

    def __init__(self,
                 local_rank: int,
                 config: KVTransferConfig,
                 hostname: str = "",
                 port_offset: int = 0,
                 library_path: Optional[str] = None) -> None:
        self.config = config
        self.rank = port_offset
        self.local_rank = local_rank
        self.device = torch.device(f"cuda:{self.local_rank}")
        flagcx_path = os.getenv('FLAGCX_PATH')
        library_path=os.path.join(flagcx_path, "build/lib/libflagcx.so")
        self.flagcx = FLAGCXLibrary(library_path)

        if not hostname:
            hostname = get_ip()
        port = self.config.kv_port + port_offset
        if port == 0:
            raise ValueError("Port cannot be 0")
        self._hostname = hostname
        self._port = port

        # Each card corresponds to a ZMQ address.
        self.zmq_address = f"{self._hostname}:{self._port}"

        # The `http_port` must be consistent with the port of OpenAI.
        self.http_address = (
            f"{self._hostname}:"
            f"{self.config.kv_connector_extra_config['http_port']}")

        # If `proxy_ip` or `proxy_port` is `""`,
        # then the ping thread will not be enabled.
        proxy_ip = self.config.get_from_extra_config("proxy_ip", "")
        proxy_port = self.config.get_from_extra_config("proxy_port", "")
        if proxy_ip == "" or proxy_port == "":
            self.proxy_address = ""
        else:
            self.proxy_address = proxy_ip + ":" + proxy_port

```

FlagCX代

https://github.com/FlagOpen/FlagCX/blob/main/plugin/interservice/flagcx_wrapper.py

FlagCX / plugin / interservice / **flagcx_wrapper.py**

Code | Blame | 420 lines (356 loc) · 16.7 KB

```python

class FLAGCXLibrary:
    exported_functions = [
        Function("flagcxHandleInit", flagcxResult_t,
                 [ctypes.POINTER(flagcxHandlerGroup_t)]),
        Function("flagcxHandleFree", flagcxResult_t,
                 [flagcxHandlerGroup_t]),
        Function("flagcxGetErrorString", ctypes.c_char_p, [flagcxResult_t]),
        Function("flagcxGetVersion", flagcxResult_t,
                  [ctypes.POINTER(ctypes.c_int)]),
        Function("flagcxGetUniqueId", flagcxResult_t,
                 [ctypes.POINTER(ctypes.POINTER(flagcxUniqueId))]),
        # Note that flagcxComm_t is a pointer type, so the first argument
        # is a pointer to a pointer
        Function("flagcxCommInitRank", flagcxResult_t, [
            ctypes.POINTER(flagcxComm_t), ctypes.c_int, ctypes.POINTER(flagcxUniqueId),
            ctypes.c_int
        ]),
        # Note that flagcxStream_t is a pointer type, so the last argument
        # is a pointer
        Function("flagcxAllReduce", flagcxResult_t, [
            buffer_type, buffer_type, ctypes.c_size_t, flagcxDataType_t,
            flagcxRedOp_t, flagcxComm_t, flagcxStream_t
        ]),

        # Note that flagcxStream_t is a pointer type, so the last argument
        # is a pointer
        Function("flagcxAllGather", flagcxResult_t, [
            buffer_type, buffer_type, ctypes.c_size_t, flagcxDataType_t,
            flagcxComm_t, flagcxStream_t
        ]),

        # Note that flagcxStream_t is a pointer type, so the last argument
        # is a pointer
        Function("flagcxReduceScatter", flagcxResult_t, [
            buffer_type, buffer_type, ctypes.c_size_t, flagcxDataType_t,
            flagcxRedOp_t, flagcxComm_t, flagcxStream_t
        ]),

        Function("flagcxSend", flagcxResult_t, [
            buffer_type, ctypes.c_size_t, flagcxDataType_t, ctypes.c_int,
            flagcxComm_t, flagcxStream_t
        ]),
```

FlagScale+FlagCX异构PD分离

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

测试机器

英伟达A800：10. 1. 15. 141

```
        GPU0  GPU1  GPU2  GPU3  GPU4  GPU5  GPU6  GPU7  NIC0  NIC1  NIC2  NIC3  NIC4  NIC5  NIC6  NIC7  NIC8  NIC9
GPU0     X    NV8   NV8   NV8   NV8   NV8   NV8   NV8   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS
GPU1    NV8    X    NV8   NV8   NV8   NV8   NV8   NV8   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS
GPU2    NV8   NV8    X    NV8   NV8   NV8   NV8   NV8   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS
GPU3    NV8   NV8   NV8    X    NV8   NV8   NV8   NV8   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS
GPU4    NV8   NV8   NV8   NV8    X    NV8   NV8   NV8   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB
GPU5    NV8   NV8   NV8   NV8   NV8    X    NV8   NV8   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB
GPU6    NV8   NV8   NV8   NV8   NV8   NV8    X    NV8   SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS
GPU7    NV8   NV8   NV8   NV8   NV8   NV8   NV8    X    SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS
NIC0    SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS    X    PIX   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS
NIC1    SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   PIX    X    SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS
NIC2    SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS    X    PIX   SYS   SYS   SYS   SYS   SYS   SYS
NIC3    SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   PIX    X    SYS   SYS   SYS   SYS   SYS   SYS
NIC4    PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS    X    PIX   SYS   SYS   SYS   SYS
NIC5    PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   PIX    X    SYS   SYS   SYS   SYS
NIC6    SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS    X    PXB   SYS   SYS
NIC7    SYS   SYS   SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   PXB    X    SYS   SYS
NIC8    SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS    X    PIX
NIC9    SYS   SYS   SYS   SYS   PXB   PXB   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   SYS   PIX    X

Legend:

  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  NV#  = Connection traversing a bonded set of # NVLinks

NIC Legend:

  NIC0: mlx5_0
  NIC1: mlx5_1
  NIC2: mlx5_101
  NIC3: mlx5_102
  NIC4: mlx5_103
  NIC5: mlx5_104
  NIC6: mlx5_105
  NIC7: mlx5_106
  NIC8: mlx5_107
  NIC9: mlx5_108
```

沐曦MC550：10. 1. 15. 67

```
mx-smi  version: 2.1.12

=================== MetaX System Management Interface Log ===================
Timestamp                                           : Fri Aug 15 17:48:26 2025

Attached GPUs                                       : 8
Device link type matrix
        GPU0    GPU1    GPU2    GPU3    GPU4    GPU5    GPU6    GPU7    Node Affinity   CPU Affinity
GPU0    X       MX      MX      MX      MX      MX      MX      MX      0               0-31
GPU1    MX      X       MX      MX      MX      MX      MX      MX      0               0-31
GPU2    MX      MX      X       MX      MX      MX      MX      MX      0               0-31
GPU3    MX      MX      MX      X       MX      MX      MX      MX      0               0-31
GPU4    MX      MX      MX      MX      X       MX      MX      MX      1               32-63
GPU5    MX      MX      MX      MX      MX      X       MX      MX      1               32-63
GPU6    MX      MX      MX      MX      MX      MX      X       MX      1               32-63
GPU7    MX      MX      MX      MX      MX      MX      MX      X       1               32-63

Legend:
  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  MX   = Connection traversing MetaXLink
  NA   = Connection type is unknown
```

A800

```
sudo docker exec -it lms-dev /bin/bash
conda activate flagscale-inference
cd /workspace/test_nvidia
```

MC550

```
sudo docker exec -it lms-dev /bin/bash
cd /workspace/test_muxi
```

FlagScale+FlagCX异构PD分离

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

☐ Step 1：在A800上启动推理服务

```
python show_disagg_xpyd.py

# show_disagg_xpyd.py

...
```

☐ Step2：在A800上启动Prefilling

```
bash show_p1.sh
```

```
# show_p1.sh
export PYTHONPATH=/workspace/FlagScale/third_party/vllm
export CUDA_VISIBLE_DEVICES=0
export VLLM_USE_V1=0
export USE_FLAGCX=1
export FLAGCX_PATH=/workspace/FlagCX_nvidia/
export FLAGCX_IB_HCA=mlx5_104
export FLAGCX_DEBUG=TRACE
export FLAGCX_DEBUG_SUBSYS=ALL
#export VLLM_CONFIGURE_LOGGING=1
#export VLLM_LOGGING_LEVEL=DEBUG
vllm serve /workspace/Qwen2-7B-Instruct \
    --host 0.0.0.0 \
    --port 20001 \
    --tensor-parallel-size 1 \
    --seed 1024 \
    --served-model-name qwen \
    --max-model-len 32768 \
    --max-num-batched-tokens 32768 \
    --max-num-seqs 5 \
    --trust-remote-code \
    --gpu-memory-utilization 0.9 \
    --kv-transfer-config \

'{"kv_connector":"P2pConnector","kv_role":"kv_producer","kv_buffer_size":"1e9","kv_port":"21001","kv_connector_extra_config":{"proxy_ip":"10.1.15.141","proxy_port":"30001","http_port":"20001","send_type":"PUT_ASYNC"}}'
```

☐ Step3：在MC550上启动Decoding

```
bash show_d1.sh
```

```
# show_d1.sh
export VLLM_USE_V1=0
export MACA_VISIBLE_DEVICES=6
export USE_FLAGCX=1
export FLAGCX_PATH=/workspace/FlagCX_muxi/
export FLAGCX_IB_HCA=mlx5_107
export FLAGCX_DEBUG=TRACE
export FLAGCX_DEBUG_SUBSYS=ALL
vllm serve /workspace/Qwen2-7B-Instruct \
    --host 0.0.0.0 \
    --port 20002 \
    --tensor-parallel-size 1 \
    --seed 1024 \
    --served-model-name qwen \
    --max-model-len 32768 \
    --max-num-batched-tokens 32768 \
    --max-num-seqs 64 \
    --trust-remote-code \
    --gpu-memory-utilization 0.98 \
    --kv-transfer-config \

'{"kv_connector":"P2pConnector","kv_role":"kv_consumer","kv_buffer_size":"1e10","kv_port":"22001","kv_connector_extra_config":{"proxy_ip":"10.1.15.141","proxy_port":"30001","http_port":"20002","send_type":"PUT_ASYNC"}}'
```

☐ Step 4：在A800上启动测试

```
cd benchmarks && bash show_test.sh
```

```
# show_test.sh
python3 benchmark_serving.py \
  --backend vllm \
  --model qwen \
  --endpoint /v1/completions \
  --dataset-name sharegpt \
  --dataset-path /workspace/benchmarks/ShareGPT_V3_unfiltered_cleaned_split.json \
  --tokenizer /workspace/Qwen2-7B-Instruct \
  --host 10.1.15.141 \
  --port 10001 \
  --num-prompts 1000
```

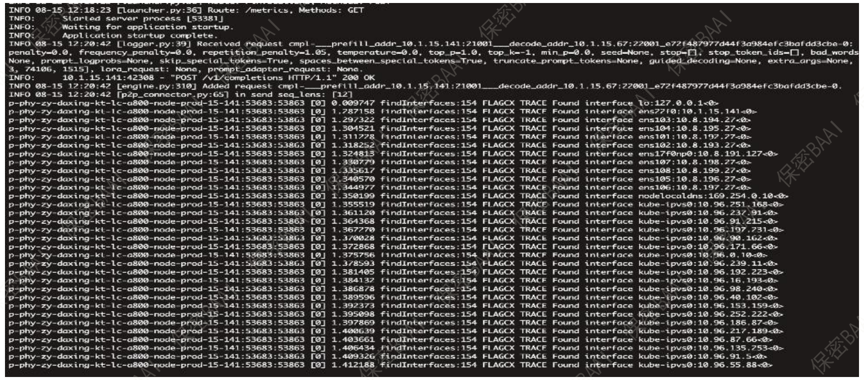
FlagScale+FlagCX异构PD分离

◻ Step 1：在A800上启动推理服务

```
python show_disagg_xpyd.py
```

◻ Step2：在A800上启动Prefilling

```
bash show_p1.sh
```

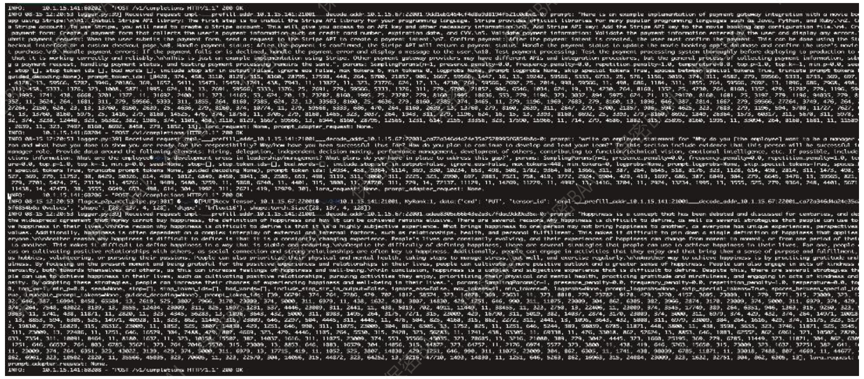


◻ Step 4：在A800上启动测试

```
cd benchmarks && bash show_test.sh
```

◻ Step3：在MC550上启动Decoding

```
bash show_d1.sh
```



FlagScale+FlagCX异构混训

北京智源人工智能研究院
BEIJING ACADEMY OF ARTIFICIAL INTELLIGENCE

PyTorch

- 通信接口支持情况（16/16）

a) send/recv/batch_isend_irecv
b) (all_)reduce/(all_)gather/(reduce_)scatter
c) (allreduce/allgather/reducescatter)_coalesced
d) all_to_all(_single)
e) broadcast
f) barrier

- 单测覆盖（16/16）

Paddle

- 通信接口支持情况（13/13）

a) (i)send/(i)recv
b) (all)reduce/(all)gather/(reduce_)scatter
c) alltoall(_single)
d) broadcast
e) barrier

- 单测覆盖（13/13）

Paddle+FlagCX GPT2 单机8卡3000迭代步对比

训练性能指标	NCCL	FlagCX
train_steps_per_second	2.86	2.85
train_samples_per_second	22.96	22.80
final_train_loss (3000steps)	6.45	6.45

FlagScale+FlagCX混训（ChipA+ChipC）

机器	模型	异构类型	加速比（vs CPU中转）
1+1	Llama3-3B	PP	9.98%
1+1	Aquila-3B	DP	4.96%
4+4	Aquila-3B	PP	7.08%
4+4	DeepSeek-16B3A	PP	3.85%

浪潮元脑集成FlagCX（ChipB+ChipG）

通信操作	通信量（Bytes）	通信带宽（GB/s）
AllReduce	128M	6.6
	1G	7.3
AllGather	128M	6.1
	1G	10.6

欢迎各位同学贡献FlagOpen开源社区！

FlagCX：*https://github.com/FlagOpen/FlagCX*

FlagScale：https://github.com/FlagOpen/FlagScale

联系我们

邮件：job@baai.ac.cn

电话：010 - 6893 3383

地址：北京市海淀区成府路150号智源大厦

智源公众号